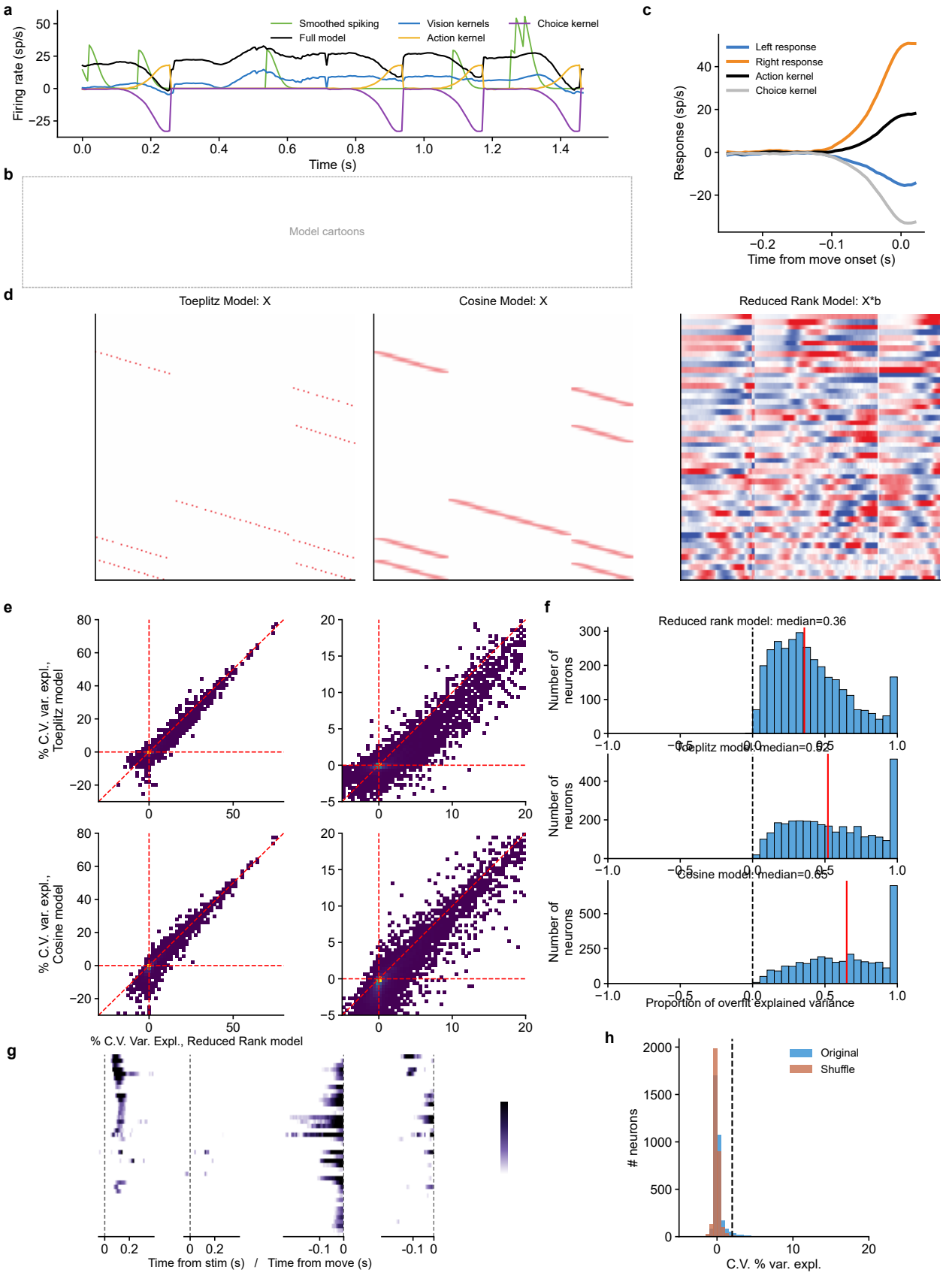

a
Smoothed spiking
Full model
Vision kernels
Action kernel
Choice kernel
Firing rate (sp/s)
Time (s)
b
Model cartoons
c
Left response
Right response
Action kernel
Choice kernel
Response (sp/s)
Time from move onset (s)
d
Toeplitz Model: X
Cosine Model: X
Reduced Rank Model: X*b
e
% C.V. var. expl., Toeplitz model
% C.V. var. expl., Cosine model
% C.V. Var. Expl., Reduced Rank model
f
Reduced rank model: median=0.36
Toeplitz model: median=0.32
Cosine model: median=0.65
Number of neurons
Proportion of overfit explained variance
g
Time from stim (s) / Time from move (s)
h
Original
Shuffle
neurons
C.V. % var. expl.